

Technical Document #921: Cloud Computing - Comprehensive Overview

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Microservices architecture structures applications as a collection of loosely coupled services. Each service runs in its own process and communicates via APIs.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. AWS EC2 and Azure VMs are popular IaaS offerings.

Multi-cloud strategies use services from multiple cloud providers. They reduce vendor lock-in and optimize for cost and performance.

Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Content delivery networks (CDNs) distribute content globally to reduce latency. CloudFront and Cloudflare are popular CDN providers.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Containerization packages applications with their dependencies.
- Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files.
- Microservices architecture structures applications as a collection of loosely coupled services.